

Terrain-Conditioned Cross-Attention for Cross-Region Avalanche Debris Mapping with Sentinel-1 SAR

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Article type: Research Article

Target journal: Journal of the Indian Society of Remote Sensing

Running title: Terrain-Conditioned Avalanche Mapping

Main-text word count: 2,084 words

Excludes abstract, tables, figure legends, and references

Figures: 2 Tables: 3 Online resources: 1

Declarations

Author Contributions

Akshay Sharma: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Visualization, Writing—Original Draft.

Lalji Prasad: Supervision, Methodology, Resources, Validation, Writing—Review and Editing.

Acknowledgements

The authors have no acknowledgements to declare.

Funding

No external grant funding was received for this study.

Competing Interests

The authors declare no competing financial or non-financial interests.

Ethics Approval and Consent

Not applicable; this study uses public remote-sensing data and involves no human participants or animals.

Consent for Publication

Not applicable.

Data and Code Availability

The AvalCD benchmark is available at <https://doi.org/10.5281/zenodo.15863589>. Code, checkpoints, the JISRS manuscript source, and verification artifacts are archived at <https://github.com/akssha74/TopoRadar-Net-AvalCD/releases/tag/v1.0.28>.

Author Agreement

Both authors approved the manuscript and agree to its submission. The manuscript has not been published and is not under consideration elsewhere.